

Topic 1: Design and Selection of Machinery Elements

Belts, Pulleys, Chains & Sprockets, Gears, Bearings, Couplings, Springs, Fasteners

Sources: Kepner, Bainer & Barger - *Principles of Farm Machinery*, and Appendix (ASAE V-belt standards); standard machine design references (Shigley; V.B. Bhandari); ASAE/IS standards.

1. POWER TRANSMISSION FUNDAMENTALS

- **Power-torque-speed:** $P \text{ (kW)} = 2\pi NT/60,000$, with N in rpm and T in N·m.
In hp: $P = 2\pi NT/4500$ (T in kgf·m).
- **Speed (velocity) ratio** of any drive = $N_1/N_2 = D_2/D_1$ (belts, neglecting slip) = T_2/T_1 (teeth, for chains/gears).
- Belt drives: friction-dependent, allow slip (overload relief), quiet, tolerate misalignment, cheap - hence dominant on farm machines (combines, threshers).
- Chain drives: positive (no slip), high load capacity, exact timing (e.g., planter/baler drives), but need lubrication and alignment.
- Gear drives: compact, positive, highest power density; used in gearboxes, final drives, rotavator trains.

2. BELT DRIVES AND PULLEYS

2.1 Flat Belts

- Transmit power by friction; pulley face is **crowned** to keep the belt tracking centrally.
- **Tension ratio (Euler/Eytelwein):** $T_1/T_2 = e^{(k\theta)}$, θ = arc of contact (rad), k = coefficient of friction.
- For rubber flat belts on steel/CI pulleys, $k \approx 0.25-0.30 \rightarrow$ tension ratio (180° arc) $\approx$ **2.2-2.6** (Kepner §4.3).
- Power: $P = (T_1 - T_2) \times v$. (Kepner eq. 4.1 in fpm/lb, units: $hp = S(T_1 - T_2)/33,000$.)
- **Centrifugal tension** $T_c = m \cdot v^2$ (m = belt mass/length) adds to both sides and reduces useful capacity; power is maximum when $T_1 = 3T_c$, giving optimum speed $v = \sqrt{(T_1/3m)}$.
- **Slip** = relative motion belt-vs-pulley (2-3% typical); **creep** = elastic shortening/lengthening as belt passes over pulleys - occurs even without overload.

2.2 V-Belts

- Wedging in the sheave groove multiplies effective friction: can work with **small arcs of contact and short centre distances** - key advantage on serpentine combine drives.
- Belt cross-section **included angle = 40°** ; sheave **groove angle $34-38^\circ$** (less than belt angle so the belt wedges; smaller groove angle on smaller sheaves compensates belt distortion).
- **Design tension ratio for V-belts = 5** ($k \approx 0.51$, includes wedging). Larger ratios risk instability due to low slack-side tension.

- A V-belt sheave and a flat pulley are “balanced” in capacity when arc of contact $\approx 130^\circ$ **on the sheave** vs **230° on the flat pulley** (equal $k\theta$) - basis of V-flat drives.
- **Arc correction factor:** ratings are for 180° arc; capacity reduced for smaller arcs (Kepner §4.4). **Service factor** accounts for load character (smooth $\rightarrow$ high shock) and expected life.
- **Belt length** (open drive): $L \approx 2C + (\pi/2)(D + d) + (D - d)^2/(4C)$; centre distance C given by standard quadratic solution.
- **Industrial sections:** A, B, C, D, E (classical); **agricultural sections (ASAE):** HA, HB, HC, HD, HE plus double (hexagonal) belts **HAA, HBB, HCC** for serpentine drives with reverse bends. Agricultural belts are designed for higher flexing life and dirt/moisture conditions.
- **Variable-speed V-belt drives:** wide belts on adjustable-pitch sheaves; speed changed on-the-go (combine ground drive) - pitch diameter range typically gives 2- to 4-fold speed variation.
- Banded (joined) belts prevent whip and turnover in multi-belt drives; V-belt matching (length grading) needed for multiple-belt drives.

2.3 Belt Tensioning

Idlers (inside/outside, tight/slack side), spring-loaded idlers as **belt clutches** on many machines (engage/disengage drive), automatic take-up. An idler on the slack side near the small pulley increases the arc of contact.

3. CHAIN DRIVES AND SPROCKETS

- Types used on farm machinery:
 - **Precision roller chain** - high speed & power (up to ~ 10 m/s); standard pitches in multiples of $1/4$ in.
 - **Detachable-link chain** (steel or malleable iron) - slow, heavily loaded conveyors/elevators, dusty conditions; cheapest, easily repaired in field.
 - **Pintle chain** - intermediate; barrel-and-pin, tolerates grit (manure spreaders).
 - **Silent (inverted-tooth) chain** - smooth, high speed, costly; rare on farm equipment.
- **Chain pitch** = distance between successive pin centres.
Chain speed $v = p \cdot T \cdot N/60$ (m/s with p in m).
- **Chordal (polygonal) action:** the chain wraps a polygon, not a circle $\rightarrow$ cyclic speed variation per tooth engagement; $variation = (1 - \cos(180^\circ/T))$ relative term; **% speed fluctuation $\approx$ decreases sharply with more teeth; negligible beyond ~ 17 – 19 teeth.** Small sprockets (< 12 teeth) give rough, noisy drives.
- Recommended small sprocket ≥ 17 **teeth** for high-speed drives; odd tooth number with even chain pitches equalizes wear.
- Sag/slack: drives run with slight slack (unlike belts, no initial tension); idler/tensioner on slack side; chain elongation limit ~ 2 – 3% (replace).
- Sprocket and roller-chain dimensions standardized (ASAE/ISO); attachment links (A-type, K-type) carry slats/paddles on conveyors - unique agricultural use.

4. GEARS

- **Law of gearing:** common normal at contact must pass through the pitch point for constant velocity ratio; involute profile satisfies this and tolerates small centre-distance change.
- **Terminology:** module $m = D/T$ (mm/tooth) - Indian/ISO standard sizing; diametral pitch (teeth/in.) in US practice; circular pitch $p_c = \pi m$. Addendum = m ; dedendum = $1.25 m$ (20° full-depth).
- **Pressure angle:** 20° full-depth is standard (14.5° obsolete); higher pressure angle $\rightarrow$ stronger tooth, fewer minimum teeth, more bearing load.
- **Minimum teeth to avoid interference** = $2/\sin^2\phi \approx 17-18$ for $\phi = 20^\circ$ (32 for 14.5°).
- **Types:** spur (parallel shafts, no axial thrust), helical (quieter, axial thrust), herringbone/double-helical (thrust cancelled), straight & spiral bevel (intersecting shafts, e.g., tractor differential crown-pinion is spiral bevel or hypoid), **hypoid** (offset axes - allows lower drive line), worm & wheel (large reduction 10–100:1 in one stage, can be self-locking, low efficiency), rack & pinion (steering).
- **Lewis beam-strength equation:** $F_t = \sigma_b \cdot b \cdot y \cdot p_c = \sigma_b \cdot b \cdot Y \cdot m$ - tooth treated as cantilever; weaker of pinion/gear (lower $\sigma \cdot Y$) is designed. Dynamic (Buckingham), wear load checks follow.
- **Velocity ratio of gear train:** simple train - intermediate (idler) gears change direction only; compound train VR = product of stage ratios; epicyclic/planetary trains give large ratios compactly (tractor final drives, planetary steering).
- Gear materials: C45/alloy steels (hardened), cast iron (low speed), nylon/plastics (light drives, seed metering gears).
- **Backlash** = play between mating tooth flanks (needed for lubrication/thermal expansion).

5. BEARINGS

- **Sliding (plain) bearings:** bush/journal (bronze, babbitt, sintered self-lubricating, wood/lignum on some ag rollers); hydrodynamic film; simple, damp shock; used at low speeds & dusty sites.
- **Rolling-element (“antifriction”) bearings:**
 - **Deep-groove ball** - radial + some axial, general purpose.
 - **Self-aligning ball / spherical roller** - misalignment tolerance (long shafts of ag machines).
 - **Cylindrical roller** - high radial capacity.
 - **Needle roller** - radial space limited (U-joints use needle bearings).
 - **Tapered roller** - combined radial + axial (wheel hubs, final drives; mounted in pairs).
 - **Thrust bearings** - axial loads.
- **Rating life:** L_{10} (a. k. a. B_{10}) = life that **90% of a bearing group will complete/exceed**: $L_{10} = (C/P)^p$ **million revolutions**, $p = 3$ for ball, $10/3$ for roller bearings; C = basic dynamic capacity, P = equivalent load.

Rating Life Equation

$$L_{10} = \left(\frac{C}{P}\right)^p$$

where:

- L_{10} (or B_{10}) = Basic rating life in **million revolutions**

- C = Basic dynamic load rating (N)
- P = Equivalent dynamic bearing load (N)
- p = Life exponent
 - $p = 3$ for **ball bearings**
 - $p = \frac{10}{3}$ for **roller bearings**

Meaning of L_{10} (B_{10}) Life

- **L_{10} life** is the number of **million revolutions** that **90% of a group of identical bearings** are expected to **complete or exceed** under specified operating conditions.
- Conversely, **10% of the bearings may fail** due to rolling-contact fatigue before reaching this life.
- It is also called the **B_{10} life** or **90% reliability life**.

Key Observations

- A **higher dynamic load rating (C)** increases bearing life.
- A **higher equivalent load (P)** significantly reduces bearing life because the relationship is exponential.
- Since roller bearings have a larger exponent ($10/3$), their life is even more sensitive to changes in load than ball bearings.

- Agricultural mountings: flanged/pillow-block housings with **triple-lip seals**, relubricatable or sealed-for-life; disc-harrow gang bearings are heavy-duty sealed units.

6. COUPLINGS AND OVERLOAD-PROTECTION DEVICES

- **Rigid couplings:** muff/sleeve, flange - perfectly aligned shafts only.
- **Flexible couplings:** pin-bush, jaw (spider), tyre - small misalignment + shock damping; **Oldham** - parallel offset; **universal joint (Hooke's)** - angular misalignment.
 - Single Hooke's joint at angle α : driven-shaft speed fluctuates each revolution (ratio between $\cos\alpha$ and $1/\cos\alpha$); fluctuation eliminated by **two joints with yokes in-phase and equal angles** (as in PTO drivelines) or CV joints (wide-angle PTO).
- **Overload safety devices :**
 - **Shear pin/bolt** - breaks at set overload (baler flywheel, rotavator, forage harvester); must be replaced after each overload.
 - **Slip (friction) clutch** - spring-loaded friction discs slip at set torque, auto-resets; protects against frequent momentary overloads (PTO drives of rotavators, balers).
 - **Jump (ratchet/cam) clutch** - cams jump with loud clatter at overload, re-engages when load drops; audible warning.
 - **Overrunning (freewheel) clutch** - transmits torque one way; lets implement inertia (e.g., rotavator, mower flywheel) overrun the tractor PTO during deceleration - a **safety must on PTO machines with high-inertia rotors**.

7. SPRINGS

- **Helical compression/extension springs:**

wire shear stress $\tau = K_w \cdot 8FC/(\pi d^2)$, where **spring index** $C = D/d$ (recommended 6–12) and **Wahl correction factor** $K_w = (4C - 1)/(4C - 4) + 0.615/C$ (accounts for curvature + direct shear).

- Deflection $\delta = 8FC^3n/(Gd)$; stiffness $k = F/\delta$; springs in series: $1/k = \Sigma 1/k_i$; in parallel: $k = \Sigma k_i$.
- **Torsion springs** (trip mechanisms), **leaf springs** (transport wheels), **Belleville washers** (clutch loading), rubber springs.
- Farm uses: spring-loaded tines (stony-ground cultivators auto-trip and reset), seed-drill tube pressure springs, clutch springs, relief-valve springs.
- Material: hard-drawn/oil-tempered spring steel (0.6–0.9% C), music wire for small springs.

8. FASTENERS, KEYS AND SHAFTS

- **ISO metric bolt property classes:** e.g., **8.8** $\rightarrow$ UTS = $8 \times 100 = 800$ MPa; yield = $0.8 \times 800 = 640$ MPa. Classes 4.6, 5.8, 8.8, 10.9, 12.9. Nuts matched by class number.
- Thread designation **M12 $\times$ 1.75** (**nominal dia** $\times$ **pitch**); coarse threads standard on farm equipment (resist stripping in dirty/rusty service); fine threads for adjustment.
- Locking: spring (split) washers, nyloc nuts, castle nut + split pin (wheel hubs), jam nuts, thread adhesives.
- **Keys:** rectangular/square sunk key (standard, $w \approx d/4$), **Woodruff** (self-aligning semicircular - tapered shafts, magnetos), feather key (axial sliding), taper key + gib head; **splines** for heavy sliding torque (PTO shaft: e.g., 6-spline 540 rpm, 21-spline 1000 rpm).
- Keys are designed in shear and crushing; keyway weakens shaft ($\sim 25\%$ strength reduction typical design allowance).
- **Shafts:** design by torsion $\tau = 16T/(\pi d^3)$; combined bending+torsion by max-shear-stress theory: equivalent torque $T_e = \sqrt{(M^2 + T^2)}$; standard transmission shaft sizes; critical (whirling) speed check for long shafts.
- **Rivets and welds** replace bolts in frames; weld strength design - Unit 8.

9. RAPID-REVISION FACT SHEET

Item	Value
Flat rubber belt k / tension ratio (180°)	0.25–0.30 / 2.2–2.6
V-belt design tension ratio	5
V-belt section included angle / groove angle	40° / 34–38°
V-flat balanced arcs	130° (sheave) vs 230° (flat pulley)
Max belt power condition	$T_1 = 3T_c; v_{opt} = \sqrt{(T_1/3m)}$
Agricultural V-belt sections	HA, HB, HC, HD, HE (+HAA, HBB double)

Min sprocket teeth (high speed)	17–19 (chordal action)
Chain speed	$v = p \cdot T \cdot N/60$
Min gear teeth, $\phi=20^\circ$	17 ($2/\sin^2\phi$)
Module	$m = D/T; p_c = \pi m$
Lewis equation	$F_t = \sigma_b \cdot b \cdot Y \cdot m$
Bearing life	$L_{10} = (C/P)^p \times 10^6 rev$; p = 3 ball, 10/3 roller
L_{10} meaning	90% survival life
Wahl factor	$(4C - 1)/(4C - 4) + 0.615/C$
Spring index C	D/d, ideally 6–12
Bolt class 8.8	UTS 800 MPa, yield 640 MPa
PTO splines	6 (540 rpm), 21 (1000 rpm)
Single U-joint speed ratio range	$\cos \alpha$ to $1/\cos \alpha$ per rev
Overrunning clutch	one-way freewheel - high-inertia PTO rotors
Slip clutch vs shear pin	auto-reset friction vs one-time breakage

10. MCQ QUESTION BANK (40 questions)

Q1. Power transmitted by a shaft is given by:

- (a) $2\pi NT/60$ W (b) $\pi NT/60$ W (c) $NT/60$ W (d) $2NT/\pi$ W

Q2. In a belt drive, the ratio T_1/T_2 equals:

- (a) $e^{(k\theta)}$ (b) $k\theta$ (c) e^θ/k (d) $\log(k\theta)$

Q3. The tension ratio commonly assumed in designing V-belt drives is:

- (a) 2 (b) 3 (c) 5 (d) 8

Q4. For rubber flat belts on steel pulleys, the tension ratio at 180° arc of contact is about:

- (a) 1.2–1.5 (b) 2.2–2.6 (c) 4–5 (d) 6–8

Q5. The included angle of a standard V-belt cross-section is:

(a) 30° (b) 34° (c) 38° (d) 40°

Q6. Sheave groove angles are kept at $34\text{--}38^\circ$, i.e., less than the belt angle, to:

(a) Reduce belt speed (b) Ensure wedging action (c) Increase slip (d) Ease belt removal

Q7. A V-belt sheave and a flat pulley are balanced in power capacity when arcs of contact are about:

(a) 180° and 180° (b) 130° and 230° (c) 230° and 130° (d) 90° and 270°

Q8. Belt power is maximum when centrifugal tension equals:

(a) T_1 (b) $T_1/2$ (c) $T_1/3$ (d) $3T_1$

Q9. Creep in belts is caused by:

(a) Overload slip (b) Alternate elastic elongation/contraction of the belt over pulleys (c) Misalignment (d) Wear

Q10. Agricultural V-belt cross-sections per ASAE standards are designated:

(a) A, B, C (b) HA, HB, HC, HD, HE (c) SPA, SPB (d) 3V, 5V, 8V

Q11. Double (hexagonal) V-belts are used for:

(a) Higher speeds (b) Serpentine drives with reverse bends (c) Vertical shafts (d) Wet conditions

Q12. Flat-belt pulleys are crowned to:

(a) Increase friction (b) Keep the belt tracking at pulley centre (c) Reduce weight (d) Balance the pulley

Q13. An idler pulley placed on the slack side near the small pulley:

(a) Reduces arc of contact (b) Increases arc of contact and capacity (c) Increases slip (d) Reduces belt life only

Q14. Variable-speed V-belt drives change speed by:

(a) Changing motor speed (b) Adjustable-pitch sheaves altering effective pitch diameter (c) Gearbox (d) Slipping clutch

Q15. Chordal (polygonal) action in chain drives causes:

(a) Chain elongation (b) Cyclic variation in chain velocity (c) Wear of pins (d) Sprocket eccentricity

Q16. To minimize chordal action, the smaller sprocket should generally have at least:

(a) 6 teeth (b) 10 teeth (c) 17 teeth (d) 30 teeth

Q17. The chain best suited to gritty, slow-speed farm conveyor service and field repair is:

(a) Silent chain (b) Precision roller chain (c) Detachable-link (steel/malleable) chain (d) Leaf chain

Q18. Chain speed (m/s) with pitch p (m), sprocket teeth T , speed N (rpm) is:

(a) $pTN/60$ (b) $pT/60N$ (c) $60pTN$ (d) $pN/60T$

Q19. Unlike belts, chain drives are operated with:

(a) High initial tension (b) A small amount of slack (c) Zero backlash (d) Crowned sprockets

Q20. Attachment links (A- and K-type) on roller chains are used in agriculture to:

(a) Join chain ends (b) Mount slats/paddles on conveyor/elevator chains (c) Reduce noise (d) Increase strength

Q21. The standard pressure angle for involute gears today is:

(a) 14.5° (b) 20° (c) 25° (d) 30°

Q22. Minimum number of teeth on a 20° full-depth pinion to avoid interference is about:

(a) 12 (b) 17 (c) 25 (d) 32

Q23. Module of a gear equals:

(a) T/D (b) D/T (c) $\pi D/T$ (d) $T/\pi D$

Q24. The gear pair connecting shafts whose axes are offset (non-intersecting, non-parallel), used in automotive/tractor drive lines, is: (a) Spiral bevel (b) Hypoid (c) Herringbone (d) Rack and pinion

Q25. A worm-and-wheel drive is preferred where:

(a) High efficiency is needed (b) Large speed reduction in one stage (and possible self-locking) is needed (c) Shafts are parallel (d) Axial thrust must be zero

Q26. In the Lewis equation $F_t = \sigma \cdot b \cdot Y \cdot m$, Y is the:

(a) Wear factor (b) Lewis form factor (tooth shape) (c) Service factor (d) Velocity factor

Q27. An idler gear in a simple gear train:

(a) Changes speed ratio (b) Changes only the direction of rotation (c) Increases torque (d) Reduces noise

Q28. L_{10} (B_{10}) bearing life is the life that:

(a) 10% of bearings complete (b) 90% of a group of bearings will complete or exceed (c) Average bearing achieves (d) The best bearing achieves

Q29. Bearing life varies with load as:

(a) $L \propto C/P$ (b) $L \propto (C/P)^3$ for ball bearings (c) $L \propto (C/P)^2$ (d) $L \propto e^{(C/P)}$

Q30. For combined heavy radial and axial loads in tractor wheel hubs, the bearing used is:

(a) Deep-groove ball (b) Needle roller (c) Tapered roller (d) Self-aligning ball

Q31. Universal-joint cross trunnions usually run in:

(a) Bush bearings (b) Needle roller bearings (c) Ball bearings (d) Tapered rollers

Q32. Speed fluctuation of a single Hooke's joint is eliminated in a PTO shaft by: (a) Higher speed (b) Two joints with in-phase yokes and equal joint angles (c) Heavier shaft (d) Lubrication

Q33. Oldham coupling accommodates:

(a) Angular misalignment (b) Parallel (lateral) shaft offset (c) Axial float only (d) Torsional vibration

Q34. The overload device that gives an audible warning and re-engages automatically is the:

(a) Shear pin (b) Jump (ratchet) clutch (c) Rigid coupling (d) Key

Q35. An overrunning (freewheel) clutch in a rotavator/PTO driveline:

(a) Limits torque (b) Allows the high-inertia rotor to overrun the PTO while slowing (c) Reverses drive (d) Increases speed

Q36. Recommended spring index ($C = D/d$) for helical springs is:

(a) 1–3 (b) 6–12 (c) 15–25 (d) 30–50

Q37. Wahl factor corrects the stress in helical springs for:

(a) Temperature (b) Curvature and direct shear effects (c) Buckling (d) Fatigue only

Q38. A bolt of property class 8.8 has minimum ultimate tensile strength and yield strength of:

(a) 800 and 640 MPa (b) 880 and 800 MPa (c) 800 and 880 MPa (d) 640 and 800 MPa

Q39. The 540-rpm tractor PTO shaft has:

(a) 21 splines (b) 6 splines (c) 8 splines (d) 10 splines

Q40. The self-aligning key seated in a semicircular shaft recess is the: (a) Feather key (b) Kennedy key (c) Woodruff key (d) Gib-head key

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	a	11	b	21	b	31	b
2	a	12	b	22	b	32	b
3	c	13	b	23	b	33	b
4	b	14	b	24	b	34	b
5	d	15	b	25	b	35	b
6	b	16	c	26	b	36	b
7	b	17	c	27	b	37	b
8	c	18	a	28	b	38	a
9	b	19	b	29	b	39	b
10	b	20	b	30	c	40	c

Selected Explanations

- **Q3/Q4:** - flat rubber belt $k = 0.25-0.30$ gives $R = 2.2-2.6$ at 180° ; V-belt design assumes $R = 5$ ($k \approx 0.51$ including wedging).
- **Q7:** equal $k\theta$ values $\rightarrow 130^\circ$ sheave $\approx 230^\circ$ flat pulley; basis of V-flat drives on threshers.
- **Q8:** $dP/dv = 0$ with $P = (T_1 - T_c - T_2')v \dots$ yields $T_c = T_1/3$.
- **Q16:** chordal velocity variation $\propto (1 - \cos 180^\circ/T)$; below 17 teeth the fluctuation and noise rise steeply.
- **Q22:** minimum teeth $= 2/\sin^2 20^\circ \approx 17.1 \rightarrow 17-18$.
- **Q28/29:** $L_{10} = (C/P)^p$ million rev; 90% reliability definition - repeatedly asked.
- **Q32:** PTO drivelines are double-Cardan arrangements; equal angles + in-phase yokes cancel the $\cos\alpha$ fluctuation.
- **Q38:** class X.Y $\rightarrow$ UTS = 100X MPa; yield = 0.Y $\times$ UTS.